

PersistentKV: Page-Aware Decode Scheduling for Long-Context LLM Serving on Commodity GPUs

Muhammad Ahmed

Abstract

Autoregressive large language model (LLM) serving is increasingly limited by key-value (KV) cache movement rather than dense matrix multiplication. Modern paged-attention systems reduce KV-cache fragmentation and mature kernels such as FlashInfer provide highly optimized native-paged decode attention. However, the best single-kernel implementation is not always the best serving schedule: low-active long-context decode can under-utilize commodity GPUs, while mixed sequence lengths introduce a tension between many exact-length launches and coarse padded batches. We present *PersistentKV*, a native block-table decode attention engine and page-aware scheduling study for grouped-query attention (GQA). PersistentKV maps work by KV-head group, executes directly over native page tables, and adds a compact workqueue schedule that executes only non-empty row-KV-head-sequence-split tasks. On an RTX 3060 with FP16, page size 16, $H_q = 32$, $H_{kv} = 8$, $d = 128$, and identical correctness tolerance against FlashInfer, a calibrated adaptive policy selects FlashInfer for small active batches, PersistentKV sequence splitting for B1 long-context steps, and PersistentKV workqueue scheduling for B8 long-context steps. With thresholds and split counts fixed on calibration traces, five held-out trace seeds improve mean synchronized wall decode-token throughput by $1.044\text{--}1.080\times$ on B8 bimodal, uniform, and Zipf-like workloads and by $1.403\times$ on a B1 bucketed trace. On the B4 bimodal boundary case, the policy avoids the PersistentKV regression by selecting FlashInfer. We also report scheduler-derived workload counters showing that workqueue scheduling reduces launch fan-out from 16.00 to 2.00 launches per decode step on held-out bimodal B8. These results identify a concrete systems niche for adaptive page-aware decode scheduling and show that work assignment, not only attention math, is a decisive serving-system variable.

1 Introduction

LLM inference systems have moved from dense, uniform tensor workloads toward stateful serving workloads dominated by KV-cache management. During prefill, attention has many query tokens and can use high-throughput matrix multiplication. During decode, each active sequence contributes only one new query token per step, while the attention kernel must stream the full prefix KV cache. The resulting arithmetic intensity is low: throughput depends on memory traffic, page-table traversal, launch scheduling, and whether enough parallel work is exposed to the GPU.

This paper studies a narrow but practically relevant question: can a native block-table decode engine improve serving latency on commodity GPUs in regimes where a production-quality paged attention kernel is already strong? The answer is not one kernel for every case. In this paper, the name *PersistentKV* refers to persistent KV-cache execution over native serving block tables: K,V pages remain in paged cache layout and the decode engine executes directly over that layout. It does not mean that the implementation uses a persistent CUDA-kernel residency model. We find that PersistentKV is useful in low-active and moderate-active long-context regimes, while FlashInfer remains the correct route for small active batches where PersistentKV’s split-merge overhead dominates. The resulting system claim is therefore a calibrated adaptive policy: choose the kernel schedule from the request state instead of forcing one attention kernel into every serving regime.

PersistentKV is evaluated on a grouped-query attention (GQA) shape, where $G = H_q/H_{kv}$ query heads share one KV head. For the Llama-style shape used in this work, $H_q = 32$, $H_{kv} = 8$, $G = 4$, and $d = 128$. The kernel assigns cooperative thread arrays (CTAs) by KV-head group and sequence split, loads K,V tiles through a native block table, and updates exact online-softmax states for all grouped query heads. A merge kernel combines split-local softmax states. We then add a workqueue route that materializes compact non-empty tasks for ragged batches, preserving one route bucket while avoiding rectangular empty split grids. The serving harness compares PersistentKV directly with FlashInfer on the same generated page-table distribution, same GQA shape, same context lengths, and same numerical tolerance.

We make three contributions.

1. We describe a native block-table GQA decode engine that exposes sequence-parallel work for low-active long-context decode while avoiding repacking K,V into contiguous tensors.
2. We add row-local sequence-length scheduling and a compact workqueue for paged decode, so split boundaries, tile prefetches, and launched tasks follow true request lengths rather than a padded rectangular route.
3. We present an adaptive serving-trace methodology with correctness checks, synchronized wall timing, CUDA-event timing, scheduler-derived workload counters, and ablations over route bucketing, raggedness, workqueue scheduling, CUDA graph replay, and fused split merging.

2 Literature Review

IO-aware attention. FlashAttention showed that exact attention can be made substantially faster by reducing high-bandwidth-memory traffic and keeping softmax statistics on chip [1]. FlashAttention-2 improved work partitioning and reduced non-matrix-multiply overheads [2]. FlashAttention-3 further exploited Hopper hardware features such as asynchronous tensor memory access and warp-group matrix multiply-accumulate [3]. These works mainly address attention kernel efficiency; our focus is the decode-serving boundary where page layout and request scheduling determine how much useful work reaches the kernel.

Paged KV-cache management. PagedAttention introduced virtual-memory style KV-cache paging for LLM serving and showed that avoiding contiguous KV allocation can substantially improve memory utilization [4]. vLLM operationalized this idea as a high-throughput serving system. TensorRT-LLM provides production kernels and runtime support for optimized LLM inference. FlashInfer provides composable high-performance inference kernels, including native paged decode attention. vAttention later argued that KV-cache paging can be exposed with lower application-level disruption through demand-paged virtual memory [6]. PersistentKV follows the native page-table line of work but focuses on low-active long-context split scheduling and row-local page-aware execution.

Continuous batching and serving traces. Orca demonstrated iteration-level scheduling for transformer serving, allowing requests with different arrival and completion times to share decode iterations [5]. Modern systems such as FasterTransformer, TensorRT-LLM, vLLM, and FlashInfer all exploit batching and specialized kernels, but the best batching policy can depend on trace structure. A single clean attention call with one sequence length cannot reveal launch-count overheads, padding waste, or low-occupancy behavior. Our methodology therefore reports both isolated native-paged baselines and continuous decode traces.

3 Methodology

3.1 Decode Attention and GQA

At decode step t , each active request has one query vector per query head and a KV cache of length N_i :

$$O_{i,h} = \sum_{j=1}^{N_i} \frac{\exp(q_{i,h}^\top k_{i,j} / \sqrt{d})}{\sum_{r=1}^{N_i} \exp(q_{i,h}^\top k_{i,r} / \sqrt{d})} v_{i,j}. \quad (1)$$

For GQA, query heads are partitioned into groups. Query heads $h \in [gH, (g+1)H)$ share the same KV head. PersistentKV maps CTAs over (request, KV head, sequence split) and computes the grouped query heads associated with that KV head inside the same work assignment. This is the implementation mapping used throughout the experiments; the paper does not separately isolate the benefit of this mapping against a CTA-per-query-head baseline.

3.2 Native Block-Table Execution

KV pages are stored as

$$K, V[\text{physical_page}, H_{kv}, \text{page_size}, d],$$

and each request has a block table mapping logical pages to physical pages. This matches paged serving layouts and avoids a repack step. PersistentKV uses a page accessor to translate token indices into physical page addresses. The attention loop processes 32-token tiles, uses asynchronous copies where available, computes tile scores, and maintains exact online-softmax state (m, ℓ, acc) in FP32.

3.3 Sequence Splitting and Row-Local Bounds

Low-active decode does not expose enough CTAs if each request and KV head owns only one CTA. PersistentKV therefore splits the sequence dimension into S disjoint ranges. Each split computes a partial online-softmax state:

$$m_s = \max_j x_j, \quad \ell_s = \sum_j \exp(x_j - m_s), \quad a_s = \sum_j \exp(x_j - m_s) v_j.$$

The merge kernel combines split states exactly:

$$m = \max_s m_s, \quad \ell = \sum_s \exp(m_s - m) \ell_s, \quad a = \sum_s \exp(m_s - m) a_s, \quad O = a/\ell.$$

For paged and bucketed execution, a key implementation detail is whether split bounds use the bucket length or the true per-row sequence length. PersistentKV uses row-local effective length:

$$N_i^* = \text{seq_lens}[i].$$

Tile count, split range, final-tile size, and prefetch size are derived from N_i^* . Empty split CTAs write neutral partial states $(m = -\infty, \ell = 0, a = 0)$ and return before loading Q or staging shared memory. This prevents coarse routing metadata from forcing padded KV reads.

3.4 Compact Workqueue Scheduling

Length-bucket scheduling has a failure mode: a ragged step with B active requests can become many PersistentKV launches, one per route length. A single coarse bucket avoids launch fan-out but creates a rectangular (B, H_{kv}, S) split grid where short rows pay for empty or inefficient split work. PersistentKV therefore adds a compact workqueue schedule. For each active row i , KV head g , and row-local split s , the scheduler emits one task

$$(\text{row} = i, \text{kvh} = g, \text{split} = s, t_{\text{begin}}, t_{\text{end}})$$

only if the tile interval is non-empty. The CUDA grid is one-dimensional over these tasks. Each task computes a partial online-softmax state and stores it in a compact slot range described by `split_offsets`. A second merge kernel performs segmented softmax-state reduction across the row-local split range. This keeps one PersistentKV route bucket for a ragged step while preserving sequence parallelism for long rows.

We also evaluated an experimental fused-merge variant in which the final split CTA for each row and KV head detects completion with an atomic counter and performs the segmented merge inside the decode kernel. This path is correct but slower on the RTX 3060 because atomic completion and in-kernel merge work cost more than the saved launch. We therefore keep the fused path as an ablation and use the two-kernel workqueue as the default.

3.5 Calibrated Adaptive Policy

The serving system exposes a manually calibrated threshold policy:

$$\text{route}(B, N_{\text{max}}) = \begin{cases} \text{PersistentKV length bucket}, & B \leq 1, \\ \text{PersistentKV workqueue}, & B \geq 8 \wedge N_{\text{max}} \geq 16\text{K}, \\ \text{FlashInfer}, & \text{otherwise.} \end{cases}$$

This policy encodes measured boundary conditions: $B \leq 1$ uses the length-bucket path, $B \geq 8$ with $N_{\text{max}} \geq 16\text{K}$ uses the workqueue path, and the remainder uses FlashInfer. B1 needs sequence splitting to expose CTAs; B8 has enough active rows to amortize the two-kernel workqueue schedule; B4 is near the threshold where merge and launch overhead dominate. The policy is simple and calibrated, not learned or self-tuning.

To avoid selecting thresholds from the same rows used for the headline table, we use a calibration/held-out protocol. The route thresholds and split counts are chosen manually on seed 20260622: split 32 for the B1 length-bucket route, split 20 for bimodal B8 workqueue, split 24 for uniform B8 workqueue, and split 28 for Zipf-like B8 workqueue. The main table reports seed 20260623 with these choices frozen. This is still not a production cost model, but it separates policy selection from reported evaluation.

3.6 Serving Harness

The harness simulates continuous decode steps. Each request has an arrival time, prompt length, decode length, physical page list, and dynamically growing current context length. FlashInfer receives one ragged native-paged batch per decode step. PersistentKV groups active requests by route length unless the single-bucket or workqueue mode is enabled. In workqueue mode, PersistentKV uses one native block-table route per step and emits only non-empty row-local split tasks. All reported comparisons use:

- the same K,V page tensors and physical page IDs;
- $H_q = 32, H_{kv} = 8, G = 4, d = 128, \text{FP16}$;
- page size 16 unless otherwise stated;
- correctness threshold $\max |e| < 2 \cdot 10^{-3}$ and $\text{mean} |e| < 3 \cdot 10^{-4}$ against FlashInfer output;
- CUDA-event timing for queued GPU work and synchronized wall timing for end-to-end step latency including Python-side planning and tensor setup.

3.7 Page-Table Generation

Each generated request first receives an exact length equal to prompt length plus decode length. The reserved logical length is then rounded up to page granularity, or to an optional bucket length in bucketed experiments. With page size P , request i owns

$$L_i = \left\lceil \frac{\text{reserved_length}_i}{P} \right\rceil$$

logical pages. Let $L = \sum_i L_i$. For a hole fraction ρ , the physical page universe has $\lceil L/(1 - \rho) \rceil$ page IDs. A seeded random permutation of this universe assigns each request a consecutive slice of physical IDs. Thus logical page order is preserved inside each request’s block table, but physical page IDs are scattered by the permutation rather than contiguous in physical memory. A 50% hole setting means that roughly half of the physical page ID universe is unassigned, so the physical address space is twice the logical page count. All methods see the same K,V tensors and the same physical page assignment. FlashInfer consumes the assignment as ragged page indices, `indptr` arrays, and last-page lengths; PersistentKV consumes the same assignment as a native block table and sequence lengths. The main serving traces use $\rho = 0$; the isolated native-paged table uses $\rho = 0.5$ to test fragmented page layouts.

Table 1: Isolated native-paged decode attention, $B = 1$, $H_q = 32$, $H_{kv} = 8$, $G = 4$, $d = 128$, FP16, page size 16, 50% holes. Lower is better.

Method	8K ms	32K ms	64K ms
FlashInfer 0.2.5	0.1194	0.4388	0.8889
vLLM 0.6.4.post1	0.1497	0.5868	1.1512
PersistentKV	0.1600	0.6169	1.2175
TensorRT-LLM 0.8 MMHA	0.7979	0.8920	1.5901
Repack + PyTorch SDPA	0.6473	2.4197	4.9840

3.8 Trace Families

We use synthetic traces because they make request structure controllable and allow each scheduling failure mode to be isolated. The *bucketed* trace cycles prompt lengths through 8K, 16K, 32K, and 64K. It models long-context interactive sessions where requests come from a small set of application-level context windows, such as code-assistant tasks, document assistants, or retrieval pipelines that use fixed maximum context budgets. The *homogeneous* trace uses a common 32K prompt length with independent decode lengths. It isolates the effect of active batch size without length heterogeneity. The *bimodal* trace mixes short and long prompts, with one long request for every three shorter requests. This is intentionally hostile to length-bucketed scheduling because most active requests have distinct current lengths after a few decode steps. The *uniform* and *Zipf* traces provide additional stress tests for broad and heavy-tailed length distributions.

These traces are not a replacement for production traffic. Their role is to separate three factors that are often conflated in one benchmark: context length, active batch size, and length raggedness. A page-aware scheduler should benefit when low active count leaves the GPU underoccupied, but it should not pay excessive launch or padding overhead when the batch is highly ragged.

3.9 Measurement Protocol

Each benchmark first constructs all request metadata and physical page assignments. K,V pages and query tensors are allocated once on the GPU. Timed steps then rebuild per-step native-paged metadata in the same style for each engine: FlashInfer receives ragged indptr, page indices, and last-page lengths; PersistentKV receives route buckets, block tables, and per-row sequence lengths. Throughput is reported in decode tokens per second,

$$\text{decode-token throughput} = \frac{\sum_t |\text{active_requests}_t|}{\text{elapsed seconds}}.$$

where each active request contributes one token per step. CUDA-event timing captures queued device work. Synchronized wall timing wraps the decode step in host wall-clock time and includes Python planning, FlashInfer planning, metadata construction, launch overhead, and synchronization. We treat CUDA-event timing as a device-work diagnostic and wall timing as the serving-facing number.

Serving-trace correctness is checked before timing by comparing PersistentKV to FlashInfer after FP32 conversion. Smoke tests separately validate smaller paged and non-paged PersistentKV shapes against a CPU FP32 reference, and the isolated native-paged baseline validates FlashInfer, vLLM, TensorRT-LLM, and PersistentKV against a PyTorch SDPA reference path. All serving tables pass the FlashInfer-equivalence tolerance; the largest reported max error is $6.104 \cdot 10^{-5}$.

4 Experiments

4.1 Experimental Setup

Experiments run on an NVIDIA RTX 3060 with 12 GB VRAM, CUDA 12.1, and PyTorch 2.5.1. The measured GPU has 28 SMs and a measured streaming read bandwidth of approximately 302 GB/s. FlashInfer 0.2.5 is used as the primary native-paged baseline. We also report vLLM 0.6.4.post1, TensorRT-LLM 0.8 MMHA, and a repack-plus-PyTorch-SDPA framework baseline for isolated native-paged tests. The TensorRT-LLM configuration uses its native MMHA fallback for the tested $G = 4$, page-16 shape.

The tested GPU is a useful target for a workshop study because it represents a commodity deployment point rather than a datacenter-only accelerator. Its 28 SMs and moderate HBM bandwidth make low-active decode underoccupancy visible: there are enough resources that a single request needs sequence splitting, but not enough bandwidth that all scheduling decisions disappear into the noise. All reported results use the same environment and should therefore be read as controlled comparisons rather than cross-machine performance claims.

4.2 Isolated Native-Paged Baselines

Table 1 verifies that the evaluation uses strong native-paged baselines. In a single-request isolated attention benchmark, FlashInfer is the fastest kernel. PersistentKV is close to vLLM but does not beat FlashInfer on this roofline-style test. This result motivates the serving-system question: the advantage must come from low-active scheduling or page-aware execution, not from claiming a universally faster single-call kernel.

Table 2: Held-out calibrated-policy serving results against FlashInfer. B1/B8 rows report mean $\pm$ std over seeds 20260623–20260627. B4 is the FlashInfer route on seed 20260623. All rows use page size 16, $H_q = 32$, $H_{kv} = 8$, $d = 128$, FP16, and pass correctness.

Trace / mode	Selected route	CUDA tok/s ratio	Wall tok/s ratio	Max $ e $	Mean $ e $
Bucketed B1	PersistentKV bucket, split 32	1.471 $\pm$ 0.037	1.403 $\pm$ 0.065	6.104 $\cdot 10^{-5}$	4.477 $\cdot 10^{-6}$
Bimodal B8	PersistentKV workqueue, split 20	1.106 $\pm$ 0.042	1.080 $\pm$ 0.050	6.104 $\cdot 10^{-5}$	4.201 $\cdot 10^{-6}$
Uniform B8	PersistentKV workqueue, split 24	1.076 $\pm$ 0.029	1.044 $\pm$ 0.022	6.104 $\cdot 10^{-5}$	3.053 $\cdot 10^{-6}$
Zipf B8	PersistentKV workqueue, split 28	1.105 $\pm$ 0.026	1.068 $\pm$ 0.028	3.052 $\cdot 10^{-5}$	2.671 $\cdot 10^{-6}$
Bimodal B4	FlashInfer baseline path	1.000	1.000	0	0

Table 3: Scheduler-derived structural counters for held-out bimodal B8 (hardware counters unavailable).

Engine	Launches/step	Decode CTA work/SM	Useful KV MB	Merge MB	Merge launches
FlashInfer	1.00	–	550.5	0.00	0.00
PersistentKV buckets	16.00	73.1	550.5	4.06	8.00
PersistentKV workqueue	2.00	45.7	550.5	2.54	1.00
Adaptive route	2.00	45.7	550.5	2.54	1.00

4.3 Main Serving Results

Table 2 reports the calibrated-policy engine over five held-out seeds after choosing thresholds and split counts on seed 20260622. The policy uses PersistentKV length buckets for B1, PersistentKV workqueue for B8 long-context steps, and FlashInfer for the B4 boundary case. This changes the claim from “PersistentKV beats FlashInfer everywhere” to the stronger serving-system claim: routing avoids known bad regimes while exploiting PersistentKV where its work decomposition helps. The largest mean wall-clock gain is 1.403 $\times$ on B1 bucketed long-context decode. Among B8 traces, mean wall decode-token throughput improves by 1.044–1.080 $\times$.

CUDA-event and wall ratios differ because wall timing also includes host planning, metadata construction, runtime launch costs, and synchronization. B1 spends some device-speed gain on split/merge overhead, while uniform B8 has a larger wall than CUDA-event ratio because the FlashInfer wall path includes planning and Python metadata work. We therefore use wall timing as the serving-facing number and CUDA events as a diagnostic.

Table 3 reports scheduler-derived workload counters for the held-out bimodal B8 run. Nsight Compute hardware counters for SM occupancy, DRAM bytes, and L2 hit rate were unavailable in the local environment because NVIDIA performance counters were permission-restricted. We therefore avoid claiming achieved occupancy or cache behavior. The scheduling mechanism is still visible: exact length buckets require 16.00 CUDA launches per decode step, while the compact workqueue requires 2.00. The workqueue also reduces merge-state traffic from 4.06 MB/step to 2.54 MB/step and merge launches from 8.00 to 1.00.

4.4 Route and Split Calibration

Sequence splitting is the primary occupancy knob. Too few splits under-utilize the GPU; too many splits increase merge overhead. B1 needs more splits because sequence parallelism is the main source of CTAs, while B8 already has active rows to amortize work. We tune split counts only on calibration traces and freeze them before reporting held-out results: split 32 for B1, and split 20/24/28 for bimodal/uniform/Zipf B8 workqueue.

This calibration is a workshop prototype, not a final routing model. The methodological point is that Table 2 reports held-out traces using fixed choices, not a per-row oracle. Table 4 shows that nearby split counts remain wall-positive but are not monotonic, so the choices should be read as calibrated operating points rather than a broad optimum.

4.5 Raggedness Ablation

The raggedness ablation is the transition from the older length-bucketed scheduler to the workqueue scheduler. Exact length buckets can turn one FlashInfer ragged batch into many PersistentKV launches. Coarse masked buckets reduce route count but still lose on bimodal traces because padded work and residual launch overhead remain. The compact workqueue restores one route while preserving sequence splitting and therefore moves held-out bimodal B8 to a wall-throughput win.

On the held-out bimodal B8 trace, exact length buckets require 16.00 launches per step, while compact workqueue requires 2.00 launches per step. This is the core raggedness result: the workqueue retains sequence splitting without multiplying route launches by the number of distinct active lengths.

4.6 Implementation Ablations

The final kernel and harness combine three implementation choices.

- **Indexed Q/O.** The extension can read Q rows and write output rows by index, avoiding per-bucket `index_select` and `index_copy_` in the persistent path.

Table 4: Split-count sensitivity on seed 20260623. Ratios are adaptive policy over FlashInfer.

Trace	Split counts	CUDA tok/s ratio	Wall tok/s ratio
Bucketed B1	28 / 32 / 36	1.606 / 1.477 / 1.375	1.379 / 1.516 / 1.282
Bimodal B8	16 / 20 / 24	1.182 / 1.181 / 1.129	1.128 / 1.136 / 1.168

Table 5: Serving GQA/MQA sweep on a smaller B8 bimodal trace, seed 20260623, 8 requests and 24 measured steps.

G	H_{kv}	CUDA tok/s ratio	Wall tok/s ratio	Max $ e $
1	32	0.489	0.504	$6.104 \cdot 10^{-5}$
4	8	1.228	1.239	$6.104 \cdot 10^{-5}$
8	4	0.263	0.276	$1.221 \cdot 10^{-4}$

- **Row-local sequence bounds.** Masked paged requests use true per-row sequence lengths for split ranges and prefetch sizes.
- **Zero-tile early exit.** Splits that own no tiles write neutral softmax states and return before staging Q or shared memory.
- **Compact workqueue.** Work items are emitted only for non-empty row-KV-head-split intervals, avoiding the launch fan-out of exact buckets.

These changes are most valuable when route metadata groups unequal lengths or when split counts are high enough that short rows would otherwise create empty work. In the mixed bimodal trace, indexed Q/O alone improved wall time but did not close the GPU gap. Row-local bounds and early exit made single-bucket ragged execution correct and reduced padded work. The compact workqueue is the first variant that makes the B8 ragged traces wall-positive on the held-out seed, with the largest held-out B8 gain appearing on bimodal traffic.

Two attempted serving optimizations did not become headline results. CUDA graph replay reduced Python launch mechanics but did not overcome the two-kernel decode-plus-merge cost on B4. Fusing decode and merge with atomic completion counters was correct, but slower on RTX 3060; the atomic and in-kernel merge overhead outweighed the saved merge launch. We keep both paths as ablations because they clarify that the remaining B4 problem is not merely host overhead.

Table 5 reports a serving GQA/MQA sweep on a smaller B8 bimodal trace that fits 12 GB VRAM for $G = 1$. The current workqueue route wins for $G = 4$ but loses for $G = 1$ and $G = 8$, so a production policy should include the GQA ratio in its routing decision.

5 Discussion

Why PersistentKV wins in low-active decode. With one active long sequence, FlashInfer’s kernel is highly optimized but the serving step exposes limited independent request-level work. PersistentKV increases parallelism by splitting the sequence while preserving exact online-softmax merging. The result is a better occupancy point on RTX 3060 for selected long-context lengths. The win appears in both CUDA-event timing and synchronized wall timing, which reduces the risk that it is merely a measurement artifact.

Why the workqueue wins at B8. Exact length bucketing exposes sequence parallelism but can require many launches. FlashInfer avoids that launch fan-out with one ragged paged batch, but its internal schedule does not explicitly target the row-local split structure used here. The compact workqueue combines the useful parts of both: one route per step and explicit sequence splitting for long rows. This is why B8 bimodal, uniform, and Zipf-like traces are wall-positive over five held-out seeds, with mean wall ratios of $1.044\text{--}1.080\times$. The gains are modest, but they hold across the tested synthetic seeds.

Why B4 routes to FlashInfer. At B4, there is enough work that no-split PersistentKV under-utilizes the GPU, but not enough work to fully amortize the second split-merge launch. CUDA graph replay and fused atomic merge both failed to solve this on RTX 3060. The adaptive system therefore routes B4 to the FlashInfer baseline path and avoids the PersistentKV regression. Improving PersistentKV itself in this regime likely requires a lower-cost merge design or a persistent cooperative kernel that performs segmented reduction without the current atomic and launch penalties.

Scope of the claim. The evidence supports a workshop-level claim: page-aware split scheduling and compact native block-table work queues can beat a strong FlashInfer baseline in selected long-context serving regimes on a commodity GPU. It does not show that PersistentKV is universally faster. A full systems paper needs a serving-runtime integration, real traces, model-level latency, more hardware targets, and current vLLM/TensorRT-LLM stacks under the same admission-control policy.

Future work. The most direct next step is a lower-overhead segmented merge for the workqueue path and an online split policy that selects different split counts for 8K, 16K, 32K, and 64K rows in the same batch. Other useful directions include page-table lookup hoisting, dynamic split selection, heterogeneous page sizes, fused RoPE/KV-cache writes, and dequantization-aware paged decode.

5.1 Threats to Validity

Synthetic traces. The traces isolate scheduling properties, but omit admission control, prefill/decode interference, tokenizer latency, network queues, memory pressure, and model-layer interactions. The results should be viewed as a controlled kernel-scheduler study.

Synthetic page allocation. The generated block tables use seeded random physical-page IDs and optional holes, so methods see identical fragmentation. This is not a production allocator: eviction, page reuse, allocator locality heuristics, and arrival-order correlations could change L2 locality and the best routing threshold.

Single hardware target. The experiments are performed on RTX 3060. Different GPUs have different SM counts, cache sizes, HBM bandwidth, Tensor Core throughput, and launch overheads. In particular, Hopper-generation GPUs provide hardware features that FlashAttention-3 and production serving systems can exploit. PersistentKV’s split policy should be retuned per GPU.

Versioned baselines. The measured FlashInfer, vLLM, and TensorRT-LLM versions are the versions compatible with the local PyTorch/CUDA environment. They are strong baselines for the present comparison, but not a claim against all future releases. The most important methodological point is that the main comparison uses identical page layout, GQA shape, context lengths, and correctness tolerance.

Held-out seed count. The main rows now use one calibration seed and five held-out seeds. This is a stronger check than a single held-out seed, but the traces are still synthetic and the B8 speedups are modest. A stronger submission should add real traffic traces and confidence intervals over more seeds and hardware.

GQA ablation scope. The serving sweep covers $G \in \{1, 4, 8\}$ on a smaller B8 trace. It shows the current workqueue route is useful for $G = 4$ but not for $G = 1$ or $G = 8$. A final version should add a query-head mapping baseline and make GQA ratio part of the production route policy.

Serving-reference correctness. Serving traces compare PersistentKV to FlashInfer, establishing equivalence to the timed production baseline rather than an independent proof of exactness for every trace. Smaller smoke tests and isolated native-paged baselines use CPU FP32 or PyTorch SDPA references; a final serving paper should add sampled SDPA checks inside the trace harness.

Kernel-only serving loop. The harness measures a decode loop, not an entire LLM server. It does not include transformer MLP layers, sampling, networking, CPU request queues, or KV-cache allocation policies beyond synthetic page assignment. A production result should integrate PersistentKV into a full serving runtime and report request-level latency distributions.

Hardware profiling permissions. Nsight Compute was available in the environment, but GPU performance counters were blocked by `ERR_NVGPUCTRPERM`. We therefore report scheduler-derived structural counters from the harness rather than hardware L2 hit rate, DRAM byte, and achieved occupancy counters. A final submission should rerun the same experiments on a machine with NVIDIA performance counters enabled.

6 Conclusion

PersistentKV studies the interface between native paged decode attention and serving scheduling. On isolated single-request native-paged attention, FlashInfer remains the strongest baseline. Under low-active long-context serving, PersistentKV’s page-aware sequence splitting improves synchronized wall decode-token throughput by $1.403 \pm 0.065 \times$ over five held-out seeds while preserving exact attention within FP16 tolerance. With compact workqueue scheduling, the same native block-table engine lets the calibrated policy reach $1.044\text{--}1.080 \times$ mean synchronized wall throughput on held-out max-active 8 bimodal, uniform, and Zipf-like traces. The method is not universal as a single kernel: workqueue B4 remains wall-negative, and the GQA sweep shows losses for $G = 1$ and $G = 8$. The calibrated policy avoids the B4 loss by selecting FlashInfer. These results make PersistentKV a credible workshop artifact: it identifies concrete regimes where page-aware scheduling can matter, provides a reproducible methodology and ablations, and exposes the next systems problem for long-context LLM serving.

References

- [1] T. Dao, D. Y. Fu, S. Ermon, A. Rudra, and C. Ré. FlashAttention: Fast and memory-efficient exact attention with IO-awareness. In *Advances in Neural Information Processing Systems*, 2022.
- [2] T. Dao. FlashAttention-2: Faster attention with better parallelism and work partitioning. arXiv preprint arXiv:2307.08691, 2023.
- [3] J. Shah, G. Bikshandi, Y. Zhang, V. Thakkar, P. Ramani, and T. Dao. FlashAttention-3: Fast and accurate attention with asynchrony and low-precision. arXiv preprint arXiv:2407.08608, 2024.
- [4] W. Kwon, Z. Li, S. Zhuang, Y. Sheng, L. Zheng, C. H. Yu, J. E. Gonzalez, H. Zhang, and I. Stoica. Efficient memory management for large language model serving with PagedAttention. In *Proceedings of the ACM Symposium on Operating Systems Principles*, 2023.
- [5] G.-I. Yu, J. S. Jeong, G.-W. Kim, S. Kim, and B.-G. Chun. Orca: A distributed serving system for Transformer-based generative models. In *USENIX Symposium on Operating Systems Design and Implementation*, 2022.

- [6] R. Prabhu, V. Janapa Reddi, and M. Zaharia. vAttention: Dynamic memory management for serving LLMs without PagedAttention. arXiv preprint arXiv:2405.04437, 2024.
- [7] FlashInfer contributors. FlashInfer: Kernel library for LLM serving. <https://github.com/flashinfer-ai/flashinfer>.
- [8] NVIDIA. TensorRT-LLM. <https://github.com/NVIDIA/TensorRT-LLM>.
- [9] vLLM contributors. vLLM: Easy, fast, and cheap LLM serving. <https://github.com/vllm-project/vllm>.